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Concept Review

Scan Match

Why Perform Scan Matching?

Scan matching allows a robot to estimate its location in space, which serves as the initial stage in the pipeline for achieving complex processes such as state estimation, simultaneous localization and mapping (SLAM), and path planning. Selecting a scan match algorithm and its parameters is often a problem of optimizing the trade-off between limits on computational resources and estimation accuracy.

What is Scan Matching?

Scan-matching, also called point-cloud registration, is the process of finding the transformation that maps one set of points to another. Scan-matching is often used in autonomous mobile robotics to make use of LiDAR data to estimate robot motion. As the robot moves around, the **current** (source) scan from its attached LiDAR sensor changes with the motion of the robot, which can be described by the transformation (i.e. the translation and rotation) that maps the current scan to a fixed **reference** (target) scan.

Scan Match Algorithms

Scan matching can be performed using various algorithms, many of which are implemented as functions in Python libraries and MATLAB toolboxes. For example, MATLAB has a function matchScans, within their Navigation Toolbox. The LIDAR Scan Match block in QUARC is based on the Normal Distributions Transform, with additional optimizations.

Iterative Closest Point (ICP) is a well-known algorithm that performs scan match by minimizing the distance between nearest-neighbour points. The distance metric can be defined in a few different ways, including Euclidean distance or root mean square distance. A main benefit of this algorithm lies in its simplicity, which makes it straightforward to understand and implement.

Normal Distributions Transform (NDT) is another scan matching algorithm that differs from ICP. NDT takes a LiDAR data scan, which spans a 2D space, and represents it as a set of probability distributions. It does so by first dividing the 2D space into a grid of equally spaced cells and then representing the location of scan points within each cell as a normal distribution. This is visualized in Figure 1 below, showing a set of scan points and the resulting probability density after applying NDT. By representing a set of discrete points as a probability density, it allows us to optimize the parameters of the that map one scan onto another.

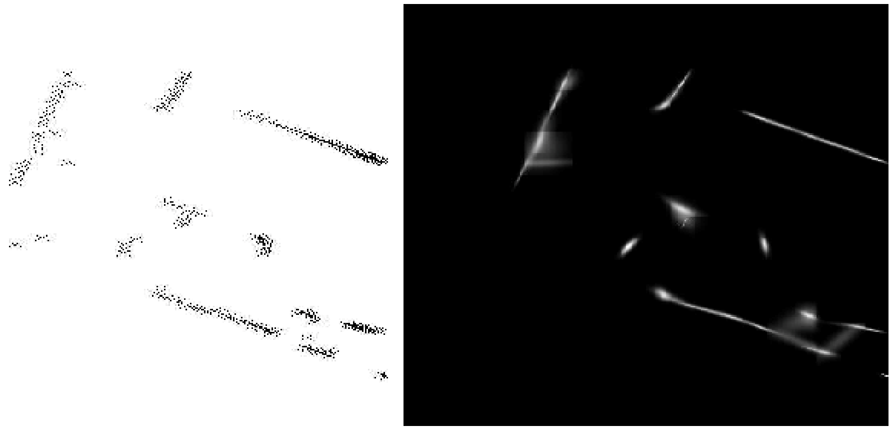

Figure 1. NDT visualization: sample scan (left) and the resulting probability density (right) [1]

[1] P. Biber, and W. Straßer, "The Normal Distributions Transform: A New Approach to Laser Scan Matching," *Proceedings 2003 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2003)*

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